Solution:

Rank by 1st referee R_{1i}	Rank by 2nd referee R_{2i}	$D_i = R_{1i} - R_{2i}$	D_i^2
1	6	-5	25
6	4	2	4
5	9	-4	16
10	8	2	4
3	1	2	4
2	2	0	0
4	3	1	1
9	10	-1	1
7	5	2	4
8	7	1	1
			$\sum_{i=1}^{n} D_i^2 = 60$

Here n = 10 and $\sum_{i=1}^{n} D_i^2 = 60$

$$\rho = 1 - \frac{6\sum_{i=1}^{n} D_i^2}{n\left(n^2-1\right)}$$

$$= 1 - \frac{6\times 60}{10\left(10^2-1\right)} = 1 - \frac{360}{10(99)} = 1 - \frac{360}{990} = 0.636$$

Interpretation: Degree of agreement between the referees 'A' and 'B' is 0.636 and they have "strong agreement" in evaluating the competitors.

Example 4.4

Calculate the Spearman's rank correlation coefficient for the following data.

Candidates	1	2	3	4	5
Marks in Tamil	75	40	52	65	60
Marks in English	25	42	35	29	33

12th_Statistics_EM_Unit_4.indd 117 1/28/2020 6:34:30 PM